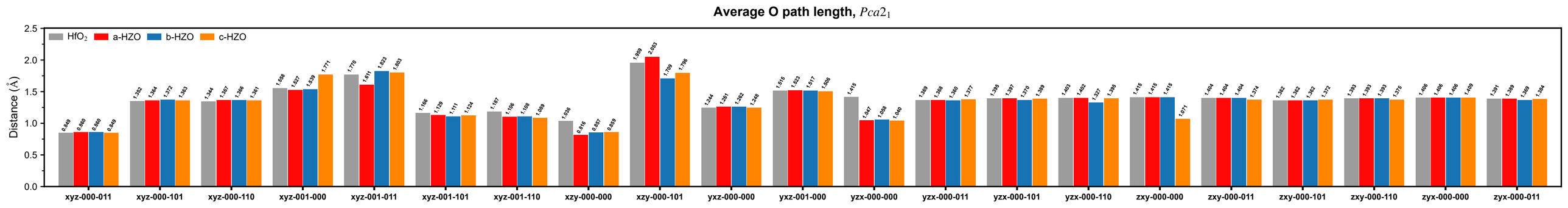
Average O path length, $Pca2_1$
Distance (Å)
2.5
2.0
1.5
1.0
0.5
0.0
HfO2
a-HZO
b-HZO
c-HZO
0.849 0.860 0.860 0.849
1.352 1.364 1.372 1.363
1.344 1.367 1.366 1.361
1.558 1.527 1.539 1.771
1.770 1.611 1.823 1.803
1.166 1.129 1.111 1.124
1.187 1.106 1.109 1.089
1.036 0.816 0.857 0.859
1.959 2.053 1.709 1.796
1.244 1.261 1.262 1.248
1.516 1.523 1.517 1.506
1.415 1.047 1.056 1.040
1.389 1.388 1.390 1.377
1.395 1.397 1.370 1.389
1.403 1.402 1.327 1.395
1.415 1.415 1.415 1.071
1.404 1.404 1.404 1.374
1.382 1.382 1.382 1.372
1.393 1.393 1.393 1.375
1.406 1.406 1.406 1.409
1.391 1.389 1.389 1.384
xyz-000-011
xyz-000-101
xyz-000-110
xyz-001-000
xyz-001-011
xyz-001-101
xyz-001-110
xzy-000-000
xzy-000-101
yxz-000-000
yxz-001-000
yzx-000-000
yzx-000-011
yzx-000-101
yzx-000-110
zxy-000-000
zxy-000-011
zxy-000-101
zxy-000-110
zyx-000-000
zyx-000-011